Question : 1

4-1 Write logic expressions and draw truth tables and analyze logic functions.

Answer:

A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

由真值表易得：

$$F_1 = A \oplus B \oplus C$$
$$F_2 = AB + AC + BC$$

效果：

F_1 为奇数有效信号校验。
F_2 为检测大于等于 2 的有效信号。
综合一起为全加器、其中 F_2 为进位。

Question : 2

4-2 Write logic expressions.

Answer:

首先由于 74LS148 输入全为 0。
所以输出 $Y_0 = Y_1 = Y_2 = 0$。
因此，输出 Y 的信号取决于 A。
即 $Y = A$。

Question : 3

4-3 Analyze logic functions.

Answer:

首先根据当前输入情况可得：

$$Y_0 = \overline{\overline{A}\,\overline{B}} \qquad Y_0 = D_0 \qquad 1 = D_0\overline{C}\,\overline{D}$$
$$Y_1 = \overline{\overline{A}\,B} \qquad Y_1 = D_1 \qquad 1 = D_1\overline{C}\,D$$
$$Y_2 = \overline{A\,\overline{B}} \qquad Y_2 = D_2 \qquad 1 = D_2C\,\overline{D}$$
$$Y_3 = \overline{A\,B} \qquad Y_3 = D_3 \qquad 1 = D_3C\,D$$

综上 $F = \overline{\overline{A}\,\overline{B}}\,\overline{C}\,\overline{D} + \overline{\overline{A}\,B}\,\overline{C}\,D + \overline{A\,\overline{B}}\,C\,\overline{D} + \overline{A\,B}\,C\,D$。
可见，当且仅当 $A = \overline{C}$ *and* $B = \overline{D}$ 时得到有效信号。
效果即为判断 A 与 C 或 B 与 D 互反。即判断 $A = \overline{C}$ 或 $B = \overline{D}$ 是否成立。
换言之，即排除 $AB = CD$ 的情况。

Question : 4

4-4 Analyze logic functions.

Answer:

观察可得，信号仅在 0（000B）以及 7（111B）时输出低电平信号。
即该逻辑电路的功能为校验三输入信号是否为同有效信号。

Question : 5

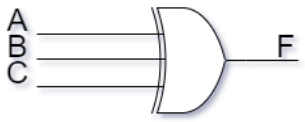
4-9 Design Circuit: Checking odd valid signal inputs

Answer:

校验奇数个有效信号，为异或门的基本功能。使用异或门即可。

$$F = A \oplus B \oplus C$$

电路图如下：



Question : 6

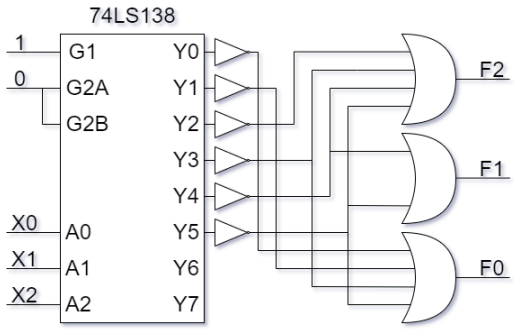
4-10 Design Circuit.

Answer:

根据题目所述，可以绘制真值表如下：

X_2	X_1	X_0	F_2	F_1	F_0
0	0	0	0	0	1
0	0	1	0	0	1
0	1	0	1	0	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	0

根据真值表，得到电路图如下：

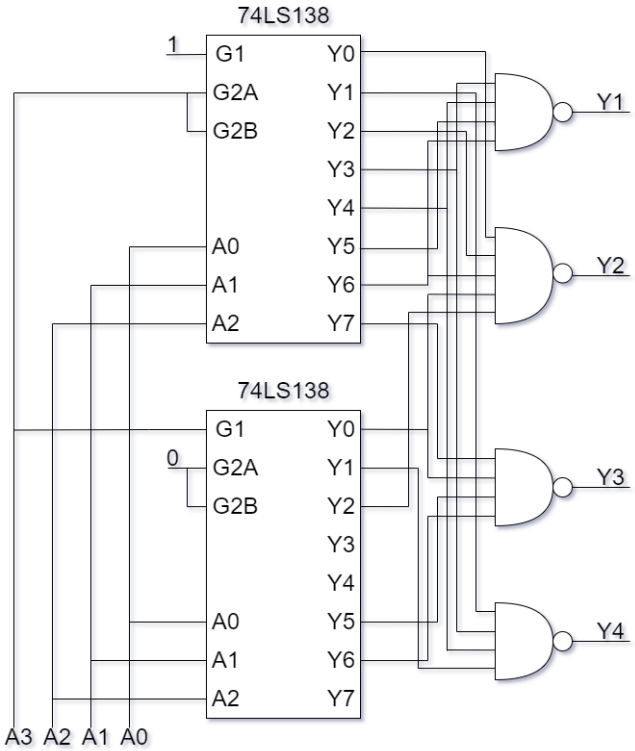


Question : 7

4-14 Design Circuit.

Answer:

由于一片 74LS138 只能表示 8 个数字，故本题需要两篇 74LS138。
根据题目所述，绘制电路图如下：



Question : 8

4-15 Coding in Verilog.

Answer:

```verilog
module half_add_sub(
    input A,
    input B,
    input X,
    output Si,
    output Ca,
    );
assign Si = A ^ B;
assign Ca = X ? (~A & B) : (A & B);
endmodule
```

Listing 1: Verilog: 15

Question : 9

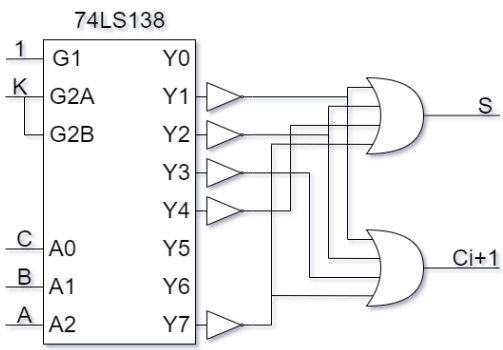
4-16 Design Circuit.

Answer:

根据题目所述，可以绘制真值表如下：

A	B	C	S	C_{i+1}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

根据真值表，可得电路图：



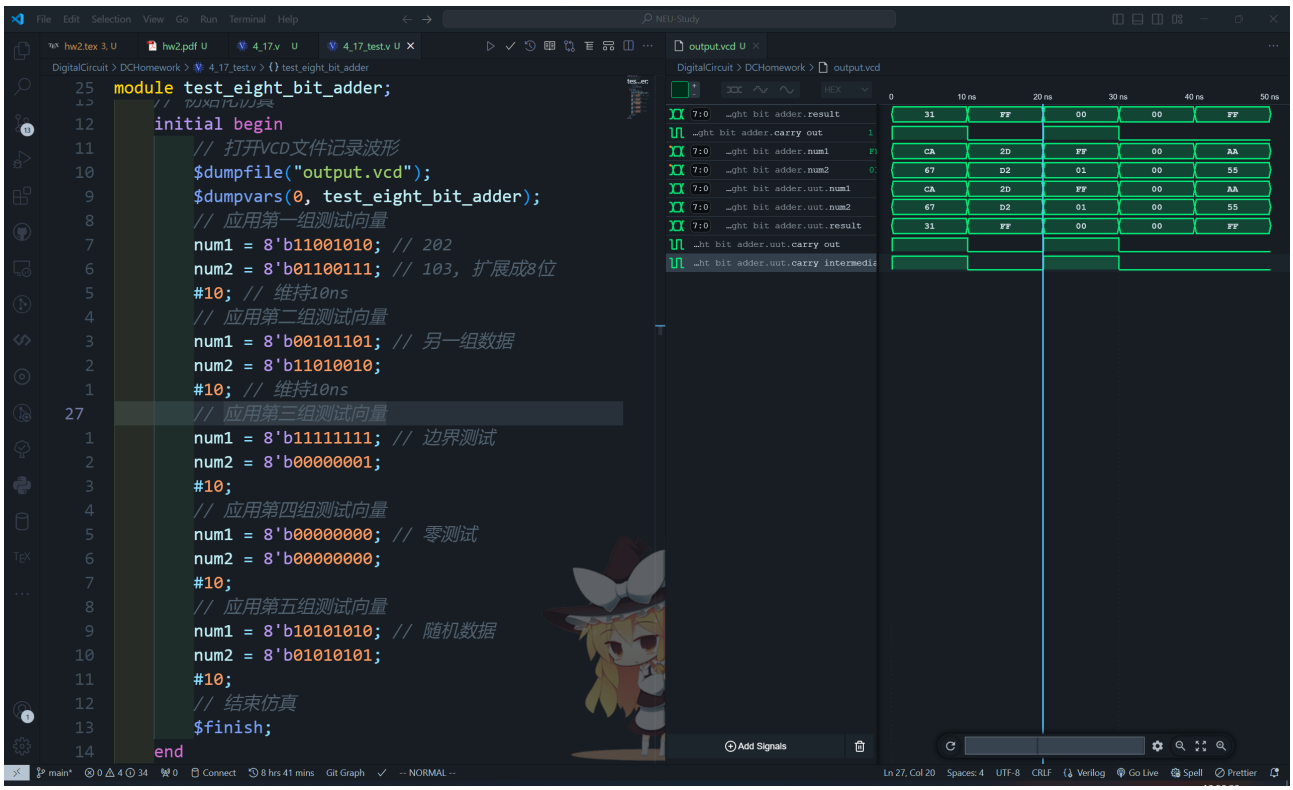
Question : 10

4-17 Coding in Verilog using 74LS238.

Answer:

```
module adder_74283(
    input [3:0] A,
    input [3:0] B,
    input Cin,
    output [3:0] Sum,
    output Cout
);
assign {Cout, Sum} = A + B + Cin;
endmodule
module eight_bit_adder(
    input [7:0] num1,
    input [7:0] num2,
    output [7:0] result,
    output carry_out
);
    wire carry_intermediate;
    adder_74283 lower_bits(
        .A(num1[3:0]),
        .B(num2[3:0]),
        .Cin(1'b0),
        .Sum(result[3:0]),
        .Cout(carry_intermediate)
    );
    adder_74283 upper_bits(
        .A(num1[7:4]),
        .B(num2[7:4]),
        .Cin(carry_intermediate),
        .Sum(result[7:4]),
        .Cout(carry_out)
    );
endmodule
```

Listing 2: Verilog: 17



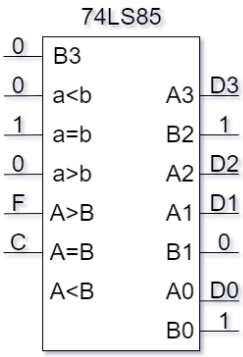
得到的波形图如下：

Question : 11

4-18 Design Circuit using 74LS85.

Answer:

易得，电路图如下：

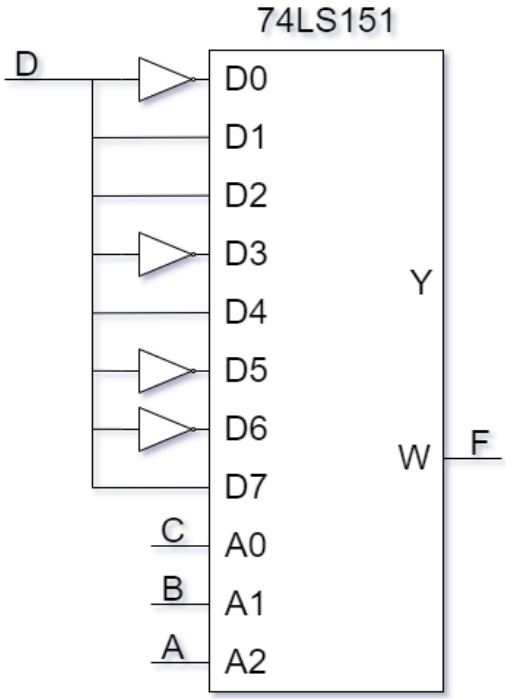


Question : 12

4-19 Design Circuit using 74LS151.

Answer:

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

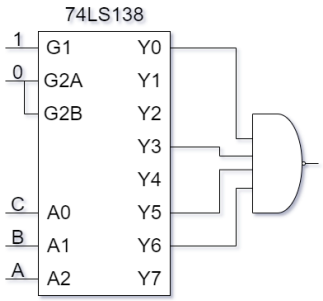


Question : 13

4-21 Reading code and draw logic circuit and draw truth tables.

Answer:

功能为偶校验。



A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Question : 14

4-22 Which number?

Answer:

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感谢您的批阅!